

Robust Short News Classification with Preprocessing and DistilBERT Midway Report

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Abstract

This project studies short news topic classification on the AG News dataset, with a focus on both standard accuracy and robustness to noisy input. I compare simple baselines, including a majority-class predictor and a TF-IDF + Logistic Regression model, against a fine-tuned DistilBERT classifier. To test robustness, I construct a perturbed test set with realistic surface-level noise such as random capitalization changes, partial punctuation removal, extra spaces, and a limited number of synthetic typos. Current results show that DistilBERT performs best on the clean test set, while both TF-IDF + Logistic Regression and DistilBERT experience small performance drops under noisy input. These findings suggest that a lightweight transformer is stronger overall for short news classification, but that surface-level perturbations still affect model reliability. In the remainder of the project, I plan to extend the evaluation with additional robustness analysis and a small adversarial test setting based on the instructor's feedback.

1 Introduction

News topic classification is a standard text classification task in natural language processing. Given an input text sequence $x = (w_1, w_2, \dots, w_n)$, the goal is to predict a label y from a fixed set of categories. In this project, I study a four-class version of the task using the labels *World*, *Sports*, *Business*, and *Sci/Tech*.

This problem is important because text classification is widely used in information retrieval, content organization, and recommendation systems. It is also a useful benchmark task because it allows clear comparison between traditional feature-based methods and pretrained transformer models. At the same time, short news texts can be difficult to classify robustly, since they often contain limited context and may be sensitive to small changes in formatting or spelling.

A simple example illustrates this difficulty. A short headline about a business event may still be understandable to a human reader after some capitalization changes, punctuation removal, or minor typos. However, a classifier may become less reliable under these small perturbations. This makes robustness an important part of evaluation, not just clean-set accuracy.

The main question of this project is whether simple preprocessing and lightweight transformer fine-tuning can improve the robustness of short-text classification. To answer this question, I compare simple baselines with DistilBERT and evaluate them on both clean and noisy versions of the AG News test set (Zhang et al., 2015).

2 Methods

I study three levels of methods.

2.1 Majority-class baseline

The simplest baseline is a majority-class classifier that always predicts the most frequent label in the training set. This provides a minimal reference point for the task.

2.2 TF-IDF + Logistic Regression

The stronger traditional baseline is a TF-IDF + Logistic Regression pipeline. The model uses unigram and bigram TF-IDF features extracted from the input text, followed by logistic regression for classification. This method is efficient, easy to implement, and provides a strong non-neural baseline for comparison.

2.3 DistilBERT

The main neural model is DistilBERT (Sanh et al., 2019), specifically `distilbert-base-uncased`. I fine-tune it for four-way classification on AG News. DistilBERT was chosen because it is smaller and faster than larger transformer models such

as BERT (Devlin et al., 2019), making it appropriate for an individual course project while still providing strong classification performance.

2.4 Noisy test construction

To evaluate robustness, I construct a perturbed version of the AG News test set. The perturbations are designed to be simple and realistic rather than destructive. They include random capitalization changes, partial punctuation removal, extra spaces, and a limited number of synthetic typos. These perturbations are intended to preserve the original topic label while making the input slightly noisier.

3 Experiments

3.1 Dataset

I use the AG News dataset as the main benchmark for this project (Zhang et al., 2015). The dataset is well suited for this task because it is widely used, manageable in size, and contains clearly defined topic labels. The full test split is used for clean and noisy evaluation. For DistilBERT training, I split the original training portion into a training set and a small validation set.

3.2 Evaluation metrics

I evaluate all models using Accuracy, Macro-F1, and Per-class F1. Accuracy measures overall classification performance, while Macro-F1 gives equal weight to each category. Per-class F1 helps identify whether some topics are more sensitive to noise than others.

3.3 Training setup

For TF-IDF + Logistic Regression, I use unigram and bigram features with a fixed feature cap and standard optimization. For DistilBERT, I fine-tune the model for two epochs with AdamW, a learning rate of 2×10^{-5} , batch size 16, and a maximum sequence length of 128. Evaluation is performed on both the clean and noisy test sets.

4 Results

Table 1 summarizes the current results on the clean and noisy test sets.

The results show that DistilBERT is the strongest model overall on the clean test set. It outperforms the TF-IDF + Logistic Regression baseline, while the majority baseline performs at chance level as expected. On the noisy test set,

both TF-IDF + Logistic Regression and DistilBERT experience small decreases in performance. This suggests that both models are somewhat sensitive to surface-level perturbations, even though DistilBERT remains the strongest overall model.

At this stage, the perturbation pipeline appears to be working as intended: the noisy set is difficult enough to produce measurable drops, but not so destructive that performance collapses. This is useful for the next stage of the project, where I plan to extend the robustness analysis further.

5 Project Plan

The main goal for the remainder of the project is to extend the current robustness evaluation beyond surface-level noise. Based on the instructor's feedback, I plan to add a small adversarial test setting in which irrelevant but misleading category cues are inserted into selected examples. The goal is to test whether classifiers rely on spurious correlations rather than true topic understanding.

I also plan to perform a brief error analysis. In particular, I will examine examples that are classified correctly on the clean test set but incorrectly on the noisy or adversarial test sets. This will help identify which perturbations are most harmful and whether different models fail in different ways.

Finally, I will organize the results into final tables and discussion sections for the final report. The final paper will include the full robustness comparison, the adversarial extension, and a clearer analysis of the tradeoff between overall performance and robustness.

References

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. 2019. Bert: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of NAACL-HLT*.
- Victor Sanh, Lysandre Debut, Julien Chaumond, and Thomas Wolf. 2019. Distilbert, a distilled version of bert: Smaller, faster, cheaper and lighter. *arXiv preprint arXiv:1910.01108*.
- Xiang Zhang, Junbo Zhao, and Yann LeCun. 2015. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems*.

Model	Clean Acc.	Noisy Acc.	! Acc. Drop	Clean Macro-F1	Noisy Macro-F1	Macro-F1 Drop
Majority Baseline	0.2500	0.2500	0.0000	0.1000	0.1000	0.0000
TF-IDF + Logistic Regression	0.9199	0.9136	0.0063	0.9197	0.9134	0.0063
DistilBERT	0.9457	0.9386	0.0071	0.9457	0.9386	0.0071

Table 1: Clean and noisy test performance for the current models.